

Sergey Didenko, Ph.D.

Engineering Leader | Principal Software Architect | Leading Engineering Teams | Embedded Systems | Medical Devices | C++ Data Platforms

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SUMMARY

Engineering leader and principal-level software architect with 20+ years delivering complex hardware/software products across embedded systems, Alexa-scale edge-and-cloud platforms, connected products, and regulated medical devices. Combines hands-on C/C++ architecture with leadership of distributed engineering teams, requirements and interface ownership, vendor management, verification and validation, and full software lifecycle execution. Builds scalable development systems and applies AI-assisted engineering to improve quality, productivity, and delivery speed.

LEADERSHIP & TECHNICAL HIGHLIGHTS

- Led and mentored engineering teams across the US, Europe, and Asia, including a six-engineer Amazon ASR core team and current process leadership with 10+ engineers.
- Designed and implemented the configurable C++ data-processing framework adopted by 11 Alexa engines across edge and cloud, cutting idea-to-production time for a new engine from 4-6 months to 3-4 weeks.
- Wrote the complete firmware from scratch in C and C++ for a connected lighting platform on ESP32/ESP32-S3, and shipped 79 device releases across two product lines.
- Led multiple concurrent regulated medical-device programs while modernizing how the software was built, and transitioned critical capability in-house for two projects through vendor coordination and knowledge-transfer.

CORE EXPERTISE

Architecture: C, C++, Python, SOLID design principles, Clean Code, Clean Architecture, configurable frameworks, systems integration, requirements decomposition, HW/SW partitioning, backend services, database systems, q/k

Embedded and connected: Embedded Linux, Yocto, QNX/RTOS, ThreadX, FreeRTOS, ESP32/ESP32-S3, STM32, ARM Cortex-M, RS-485, I2C, UART, CAN, MQTT, Wi-Fi, BLE and Bluetooth Classic, RF remote control, OTA firmware update, JTAG/SWD programming and debugging, sensor and peripheral integration, multithreaded/parallel programming

Medical and leadership: IEC 62304 Class B/C, ISO 13485, ISO 14971, IEC 60601, design controls, V&V, team leadership, mentoring, vendor management, recruiting, AI-assisted development

PROFESSIONAL EXPERIENCE

Verathon | Senior Software Engineer | Jul 2026-Present

- Drive adoption of software-development best practices across the Airway device portfolio and 200+ repositories, including SOLID design principles, Clean Code, and Clean Architecture.
- Establish scalable development and verification processes with a 10+ engineer team, introducing a structured Agile/Kanban cadence, and lead adoption of AI-assisted engineering.
- Streamline production-grade internal tools in close collaboration with IT, aligning engineering workflows with enterprise systems and support requirements.

GloMove | AI-Assisted Development & Verification | May-Jul 2026

- Established an AI-assisted development and verification process around a custom Claude Code engineering harness, enabling a two-person team to execute the complete idea-to-production cycle.
- Developed and released shared PowerUnit and Light firmware on ESP32/ESP32-S3, delivering 49 Light and 30 PowerUnit releases through the OTA update path with every release verified on real hardware.
- Automated integration, release, and end-to-end testing into a CI pipeline that exercised physical devices ahead of every firmware release, and streamed device telemetry over MQTT into Grafana dashboards.

Olympus Corporation of the Americas | Sr. Principal Software Engineer | Jul 2023-May 2026

- Led software engineering across multiple regulated medical-device programs spanning embedded connectivity, Linux platforms, video processing using i.MX8M and Cortex-M4/M7, and STM32/ThreadX fluid-control subsystems.
- Defined and drove the embedded software architecture for STM32/ThreadX fluid-control and Cortex-M4/M7 subsystems, owning HW/SW partitioning, interface definition, and integration strategy under IEC 62304 Class B/C.
- Translated system needs into software requirements, subsystem architecture, HW/SW partitioning, interfaces, and integration strategies with systems, hardware, QA/RA, operations, and manufacturing teams.
- Led verification and validation activities including protocol creation, results review, traceability improvements, and release readiness under IEC 62304 Class B/C processes; architected device-health monitoring via cross-processor watchdog timers.
- Managed external partners through late-phase delivery and knowledge-transfer, eliminating external-vendor reliance for two projects and establishing durable in-house ownership.
- Established development workflows, engineering standards, design-control documentation, CI/CD pipelines, repositories and coverage guardrails, technical interviewing, AI-enabled practices, Working Backwards, and the Build & Learn community.

GloMove | Founder & Chief Systems Architect, personal venture built evenings and weekends | Dec 2021-Aug 2023

- Wrote the complete device firmware from scratch in C and C++ for an ESP32/ESP32-S3 connected lighting platform, owning the architecture, RTOS threading model, task priorities, sensor handling, wire protocols, and lighting effects across power-unit and light devices.

- Integrated five devices on a shared I2C bus - an AHT20 temperature/humidity sensor, two INA3221 power monitors, and a DS3231 battery-backed RTC with its AT24C32 EEPROM - drove the RS-485 link between power units and lights, and programmed and debugged custom PCBAs over JTAG/SWD alongside the hardware engineer.
- Built the OTA firmware-update path end to end, and implemented BLE provisioning to pair power units with the mobile app before consolidating on Wi-Fi-only provisioning.
- Defined end-to-end architecture for the distributed platform using custom PCBAs, motion, ambient-light and capacitive-touch sensing, RF remote control, MQTT cloud connectivity with a telemetry dashboard, and mobile-connected control.
- Built and led a cross-functional team of hardware, embedded-software, and backend engineers across architecture, prototyping, manufacturing strategy, and supplier selection.

Appian Corporation | Lead Software Engineer | Dec 2021-Jan 2023

- Helped evolve backend database services and developed practical experience with q/k programming and database-oriented systems.

Amazon | Alexa | ASR Engine | Jul 2017-Sep 2021

- Senior Software Development Engineer; led a six-engineer ASR core team responsible for foundational engine functionality, legacy support, new core libraries, and migration of 10+ teams.
- Designed and implemented a configurable C++ data-processing framework adopted by 11 Alexa engines from tiny single-core devices to multi-core cloud machines, improving onboarding and time to production by 4-5x.
- Drove modularization of a monolith with 60+ libraries, 50+ binaries, less than 40% coverage, and 50+ annual contributors; achieved 95%+ coverage on decoupled modules with clearer dependencies and reduced blast radius.
- Created documentation, templates, release patterns, and migration paths that improved usability, cross-team adoption, and framework scalability across the Alexa ecosystem.
- Reduced idea-to-production time for a new ASR engine from 4-6 months to 3-4 weeks, simplified monitoring and debugging, led root-cause analysis for complex distributed failures, and drove security certification for 30+ components.

Amazon | Alexa | TTS Engine and Kindle | Aug 2015-Jul 2017

- Software Development Engineer; added the first DNN-model support to Alexa TTS and released seven engine versions, including one used for the Amazon Polly launch.
- Led design and implementation of a runtime add-on system that grew from 2 to 140 add-ons over four years and helped resolve critical launch-time runtime issues.
- Led TTS runtime-database modularization, reducing coupling and engine footprint while improving scalability across locales and languages.
- Contributed features and defect fixes to the Kindle Bird's Eye View reading experience.

Samsung Electronics, HQ | Senior Software Engineer / Project Lead | Oct 2005-Mar 2015

- Progressed through Software Engineer, Remote Team Manager, XOA-E SDK Project Leader, and Senior Software Engineer in the Software Architecture Lab.
- Led SDK, common-component, simulator, and developer-experience work for the XOA-E printing platform, including requirements clarification, risk management, V&V coordination, tools, samples, guides, and releases; defined the C++/Java APIs that opened the platform to third-party developers.
- Delivered firmware and platform software that shipped to high-volume manufacturing across Samsung MFP product lines, working with production and quality teams through launch.
- Managed a remote engineering team, including requirements clarification, time and resource planning, and delivery of Scan to Email through USB, SMB, and FTP, direct-printing, and job-flow capabilities.
- Designed multi-partition formatting, file defragmentation, and Immediate and On-Demand Image Overwriting algorithms for Samsung MFP platforms, earning an Excellence Award.
- Built image-transfer pipelines for TIFF, BMP, JPEG, GIF, PNG, and PDF variants, including secure and digitally signed formats.

Incom Ltd | Software Developer | Mar 2000-Jun 2005

- Developed software for integrated vehicle-location systems, including navigation-data encoding/decoding and optimized scheduling of transmissions between central systems and vehicles over radio and satellite links.
- Integrated MapInfo geo-information capabilities with data-transfer systems for real-time navigation-data processing and analysis.
- Clarified requirements for a USAID-supported meteorological data-collection system and contributed to scientific research.

STANDARDS & QUALITY SYSTEMS

Working knowledge of ISO 13485, ISO 14971, IEC 62304 Class B/C, IEC 60601, FDA QSR / 21 CFR Part 820, design controls, DHF-aligned documentation, traceability, verification and validation, risk review, threat-modeling collaboration, and regulated software lifecycle practices.

EDUCATION & CREDENTIALS

Doctor of Philosophy (Ph.D.) in Computer Science - Tomsk Polytechnic University, 2004 | Master of Science (M.S.) in Computer Science - Tomsk Polytechnic University, 2001

IEC 62304 Training (Developing Medical Device Software) | QNX Momentics IDE | Realtime Programming for QNX Neutrino RTOS | Writing Drivers for Neutrino | Designing RESTful APIs

Volunteer: C++ coach, FIRST Robotics team, Westborough High School - CAN bus control of motors and actuators.